

Role of data in AI systems.

→ AI systems are built on top of data.

→ Without data, AI can't predict & generate the response.

Why data is required?

→ Training

→ Inference

↳ Using a pre-trained model to generate the response.

Why efficient storage is critical?

⇒ If data is not stored properly in AI systems, it will become slow, unreliable & difficult to manage.

Database.

→ An organised system for storing, managing & retrieving data.

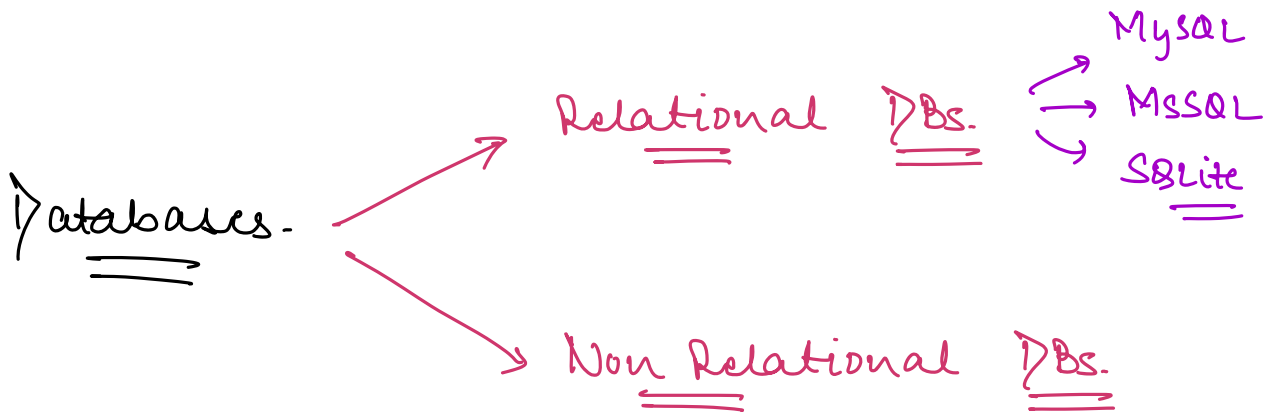
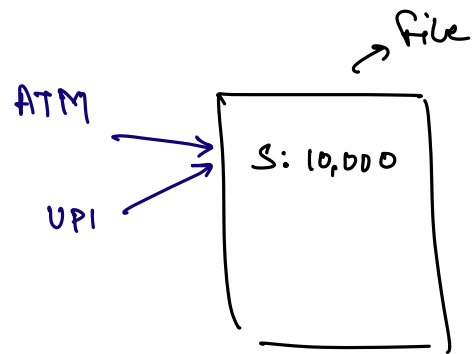
→ Create
Read
Update
Delete

File System

↳ Txt | Pdf | CSV | Excel | ...

Problems with file storage system.

- Poor scalability.
- Data inconsistency.
- Difficult retrieval.
- Concurrency issue



Relational DB.

- Stores data in the form of tables.
- Each table represents a specific entity, and relationships between the entities are maintained by keys.

Ex: MySQL
PostgreSQL
SQLite
MSSQL.

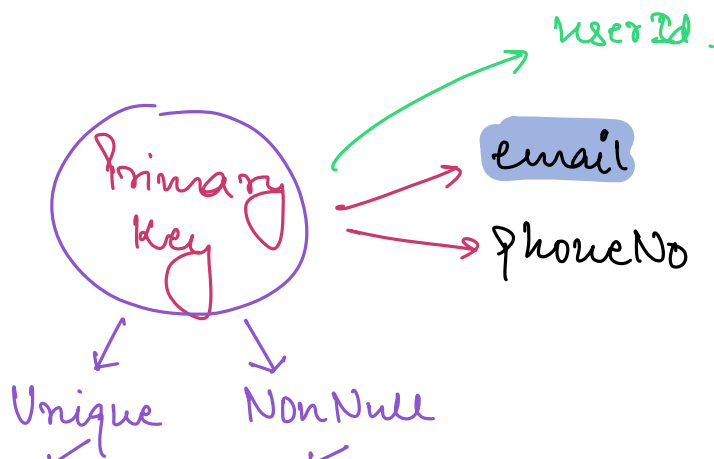
Students

<u>userId</u>	email	name	phone	—
1	abc@gmail.com	Shourya	123456	—
2	xy2@—	Nisarg	7856	—
3				
4				
5				

Columns. ⇒ Attributes.

Record.

1 Row
1 Row



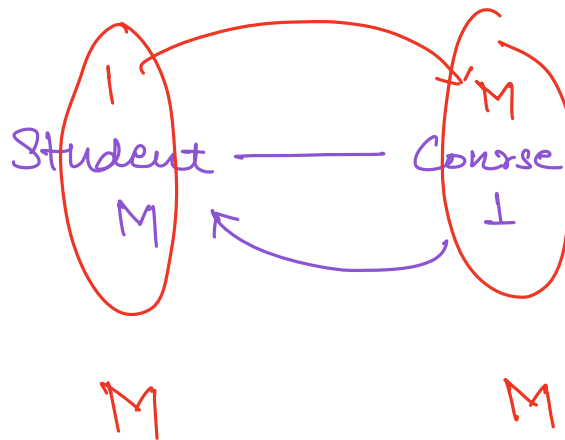
courses

↳ New Entity

Courses

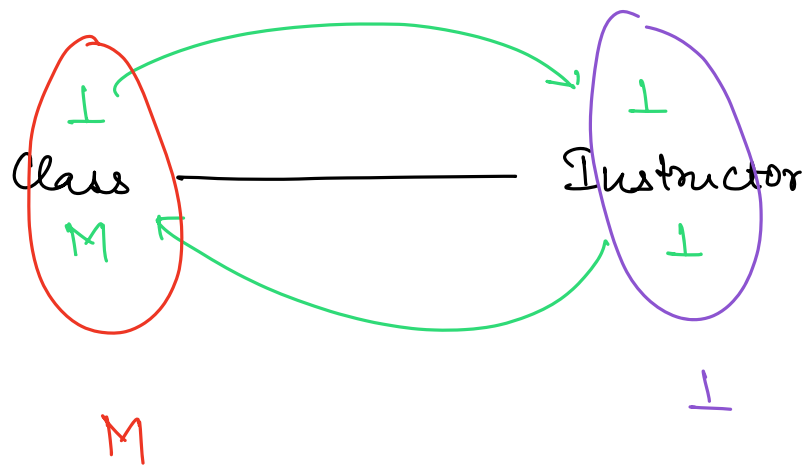
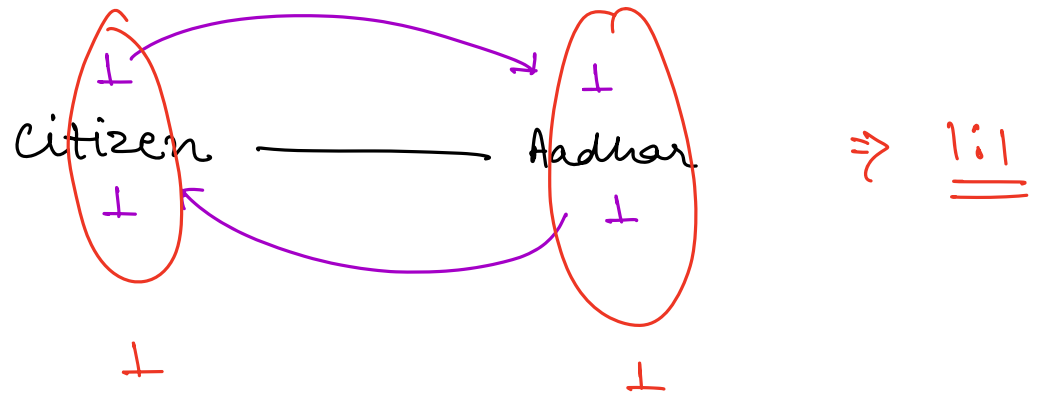
id	name	duration
1	Python	3
2	Gen AI	6
3	LLM	2
4	ML	12
=	=	=

Relationship b/w Student & Course

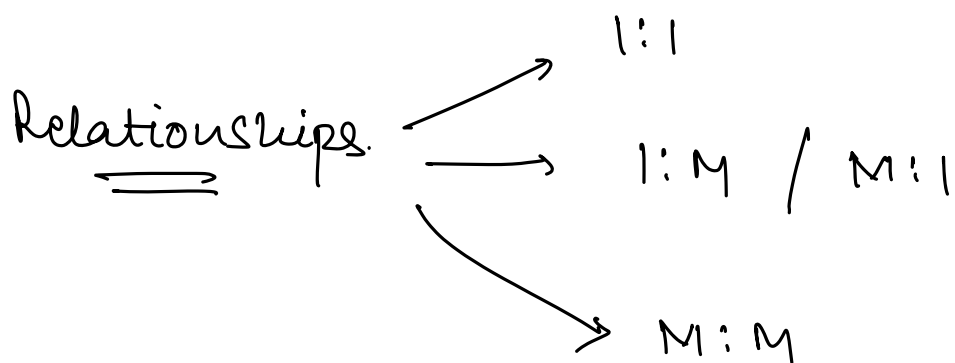


>1 ⇒ Many.
0

⇒ M:M



⇒ (M:1)



Classes

id	title	-----	instructor_id
101	DB	_____	⑦
200	AI	_____	⑧

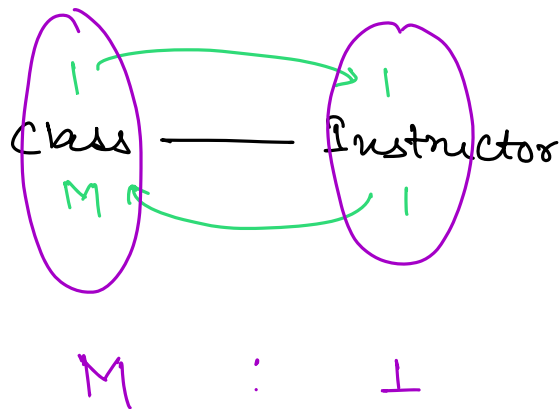
C - I

M:1

Foreign
Key.

instructors

id	name	email	list<classes>
⑦	Deepak	_____	



⇒ Id of ① side on ① side.

Primary Key

→ Unique

→ Can't be NULL.

Foreign Key.

→ Column that refers to the PK of other table.

citizen & Aadhar ⇒ (1:1) ⇒ Id of 1 side on other side.

PK

citizens

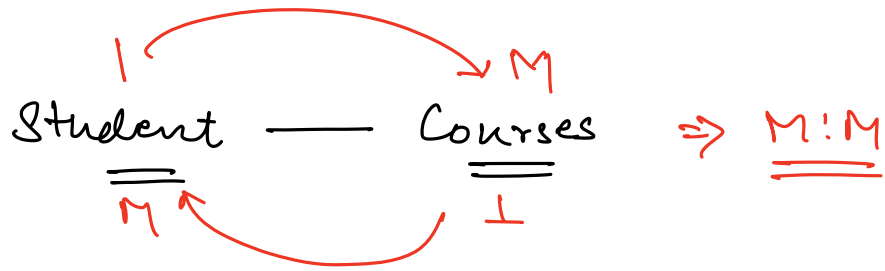
id	name	address	aadhar_id
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PK

aadhar

id			citizen_id
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⇒ (M:M)



Students

id	name	email	Courses
10	Shourya	—	[-----] X

Courses

id	name	duration	Students
1	AI	6	[-----] X

Mapping / Junction table

Student_Courses

Student_id	Course_id
10	107
10	200
10	400
=	=
20	107
50	107

Relationships.

$\Rightarrow 1:1 \Rightarrow$ Id of any side on other side.

$\Rightarrow 1:M / M:1 \Rightarrow$ Id of 1 side on M side.

$\Rightarrow M:M \Rightarrow$ Mapping / Junction table.